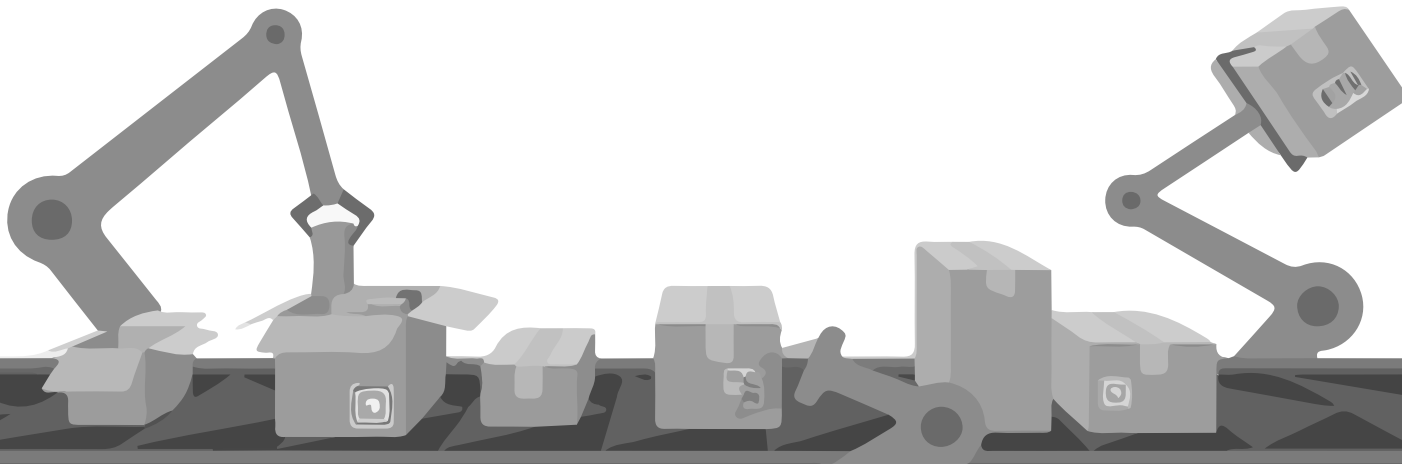
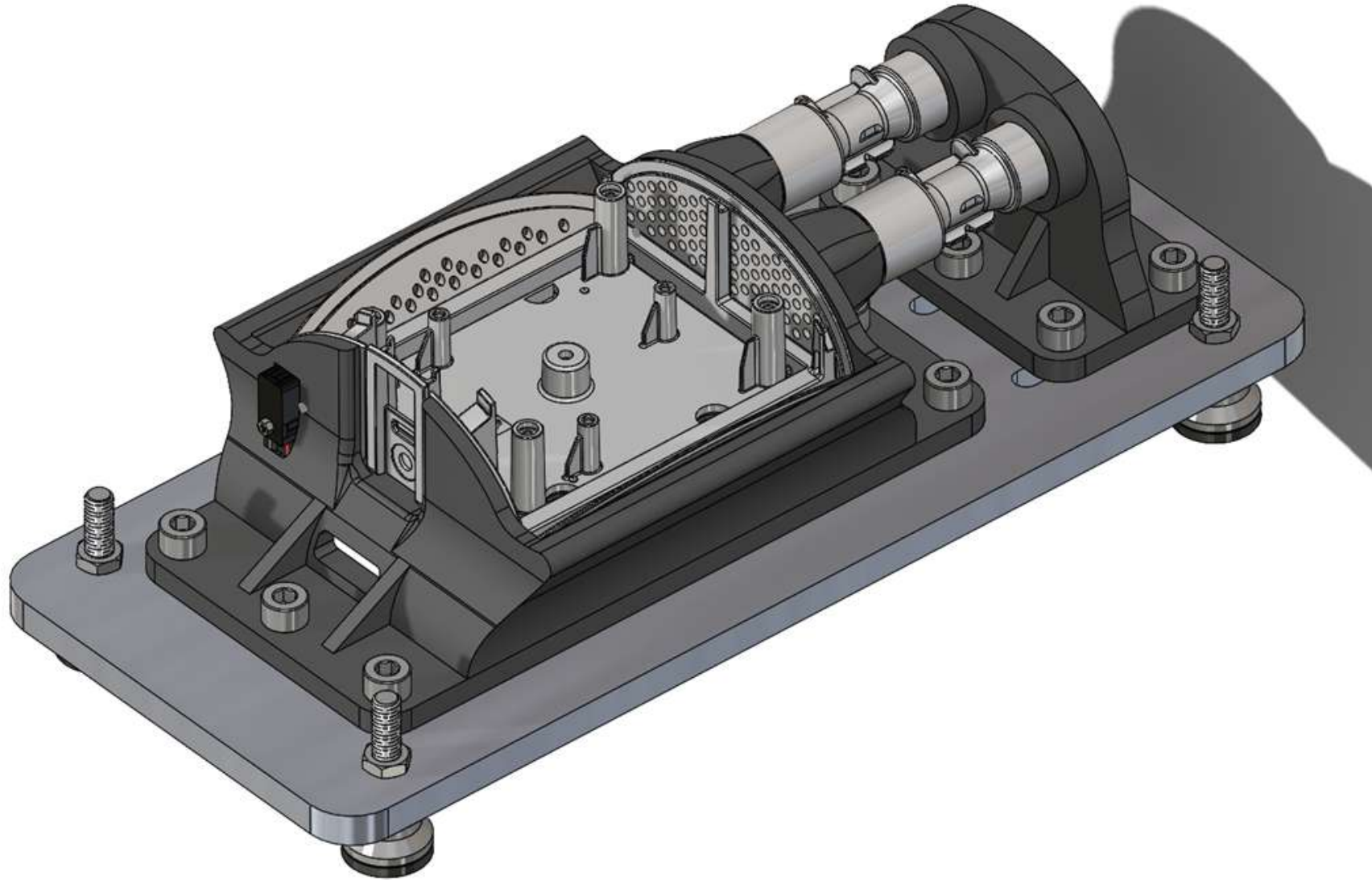


Process/Industrial Engineering

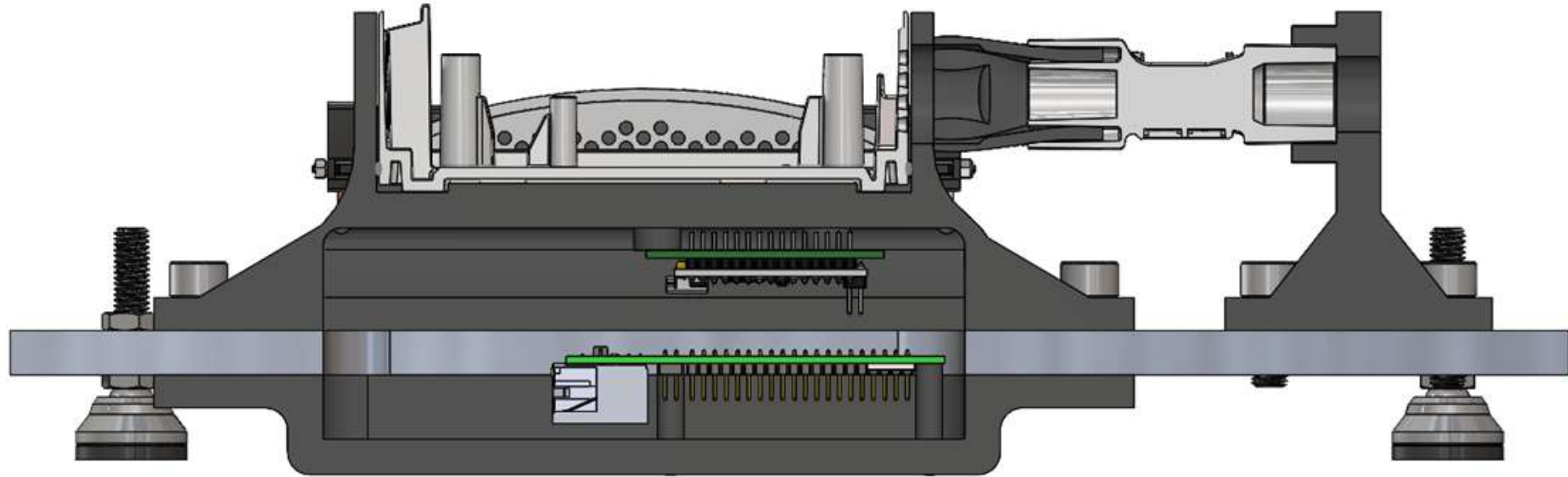
Spenser Millburn² Portfolio 2019-2022



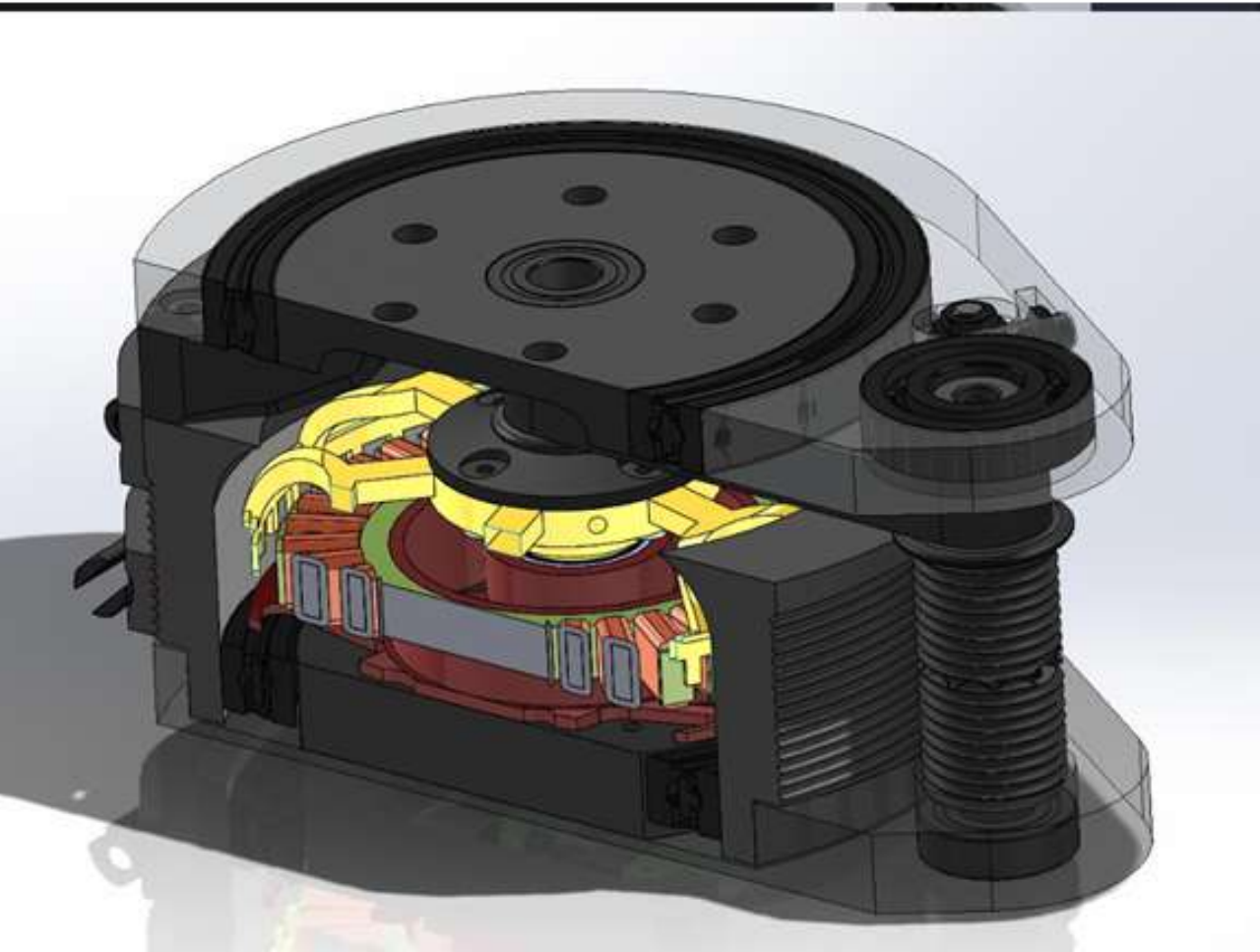
3 U R G X F W L R Q) L [W X U H
+ L J K 7 K U R X J K S X W \$ L U I O R Z 9



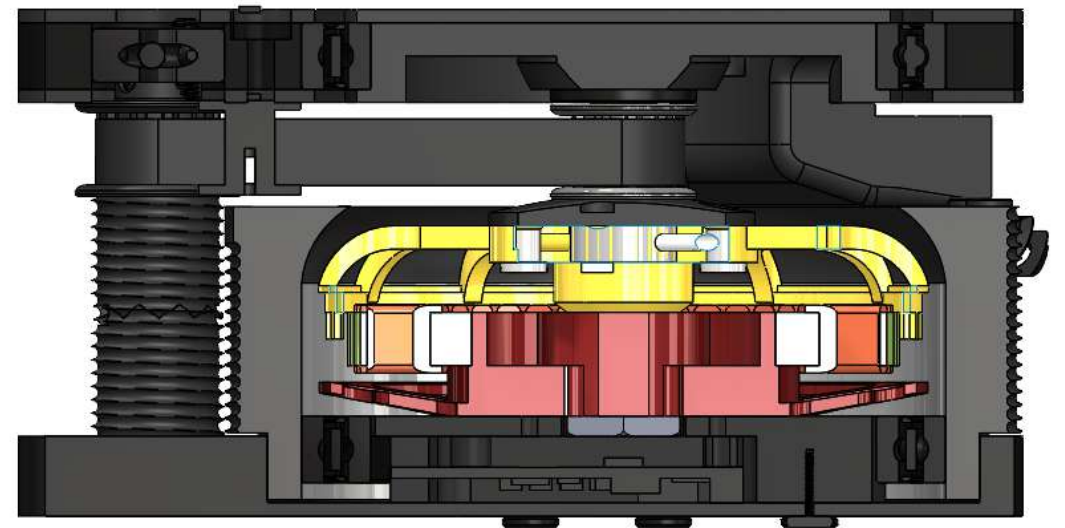
3 U R G X F W L R Q) L [W X U H
+ L J K 7 K U R X J K S X W \$ L U I O R Z 9



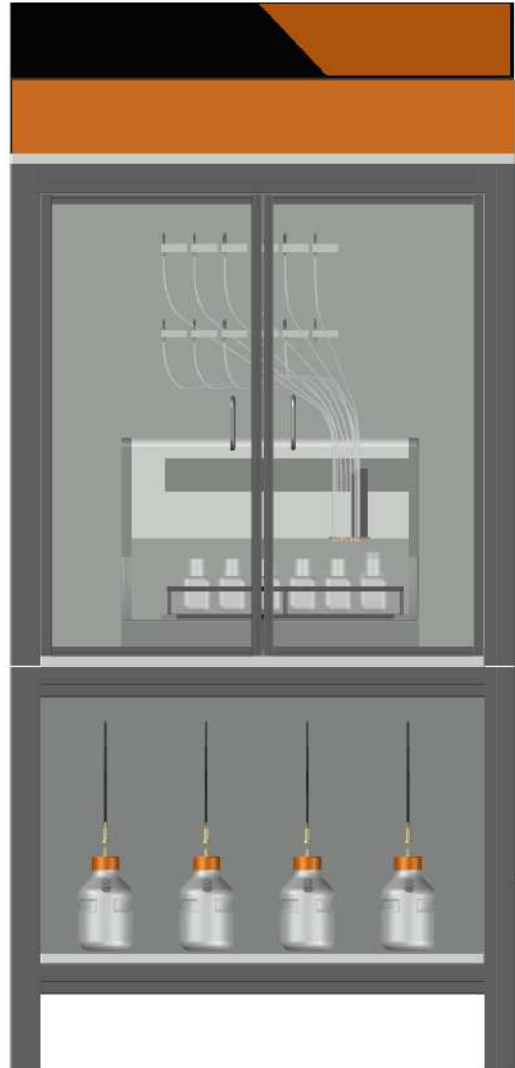
4XDGUXSHG 5RERW &DSVWDQ 'ULYH



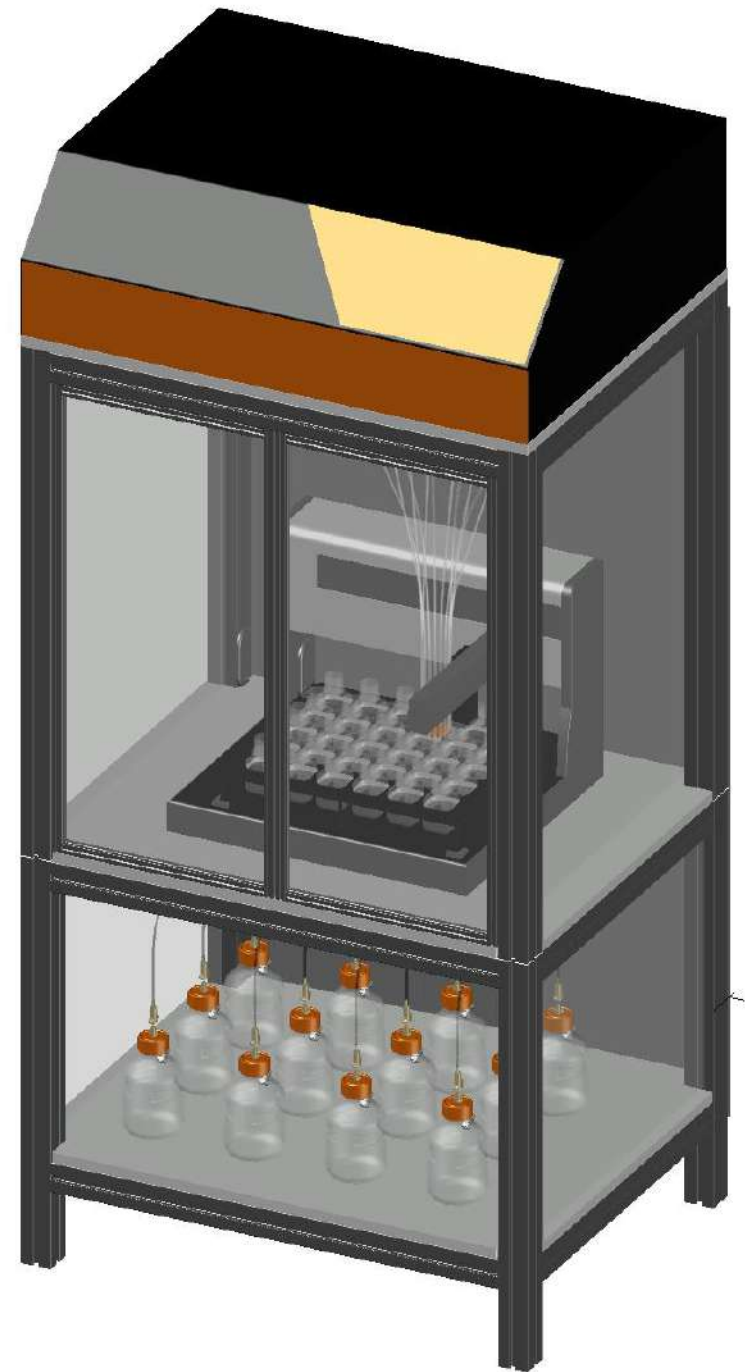
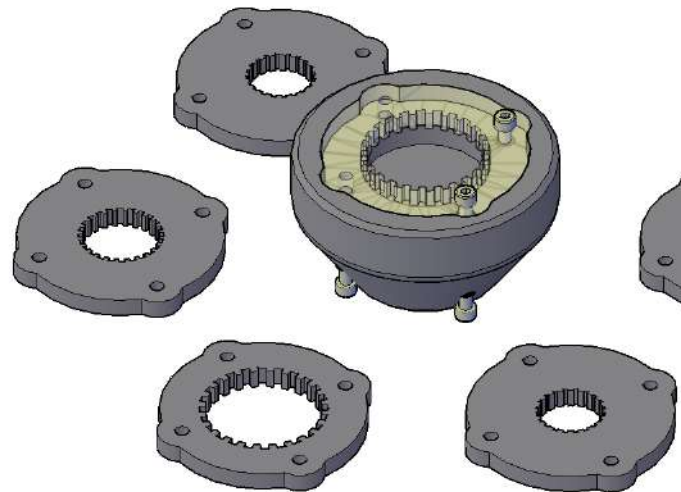
- Cable driven gear reduction
- Brushless motor successfully mounted internally in the large drum.
- Less than \$200 in parts
- Parametric model in process for easy reconfiguration of gear ratios.



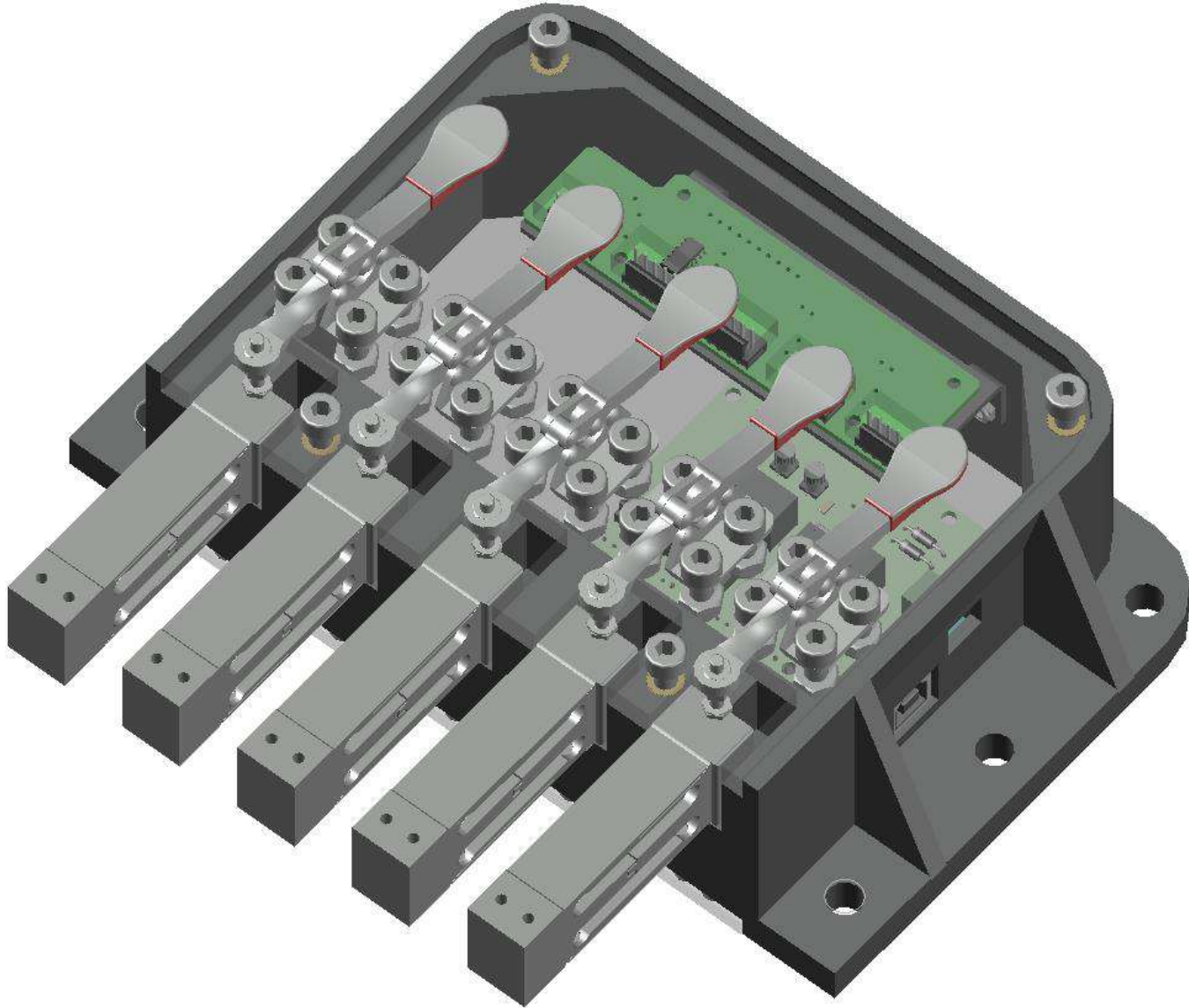
Reagent Lab Automation: Liquid Handler/Capper



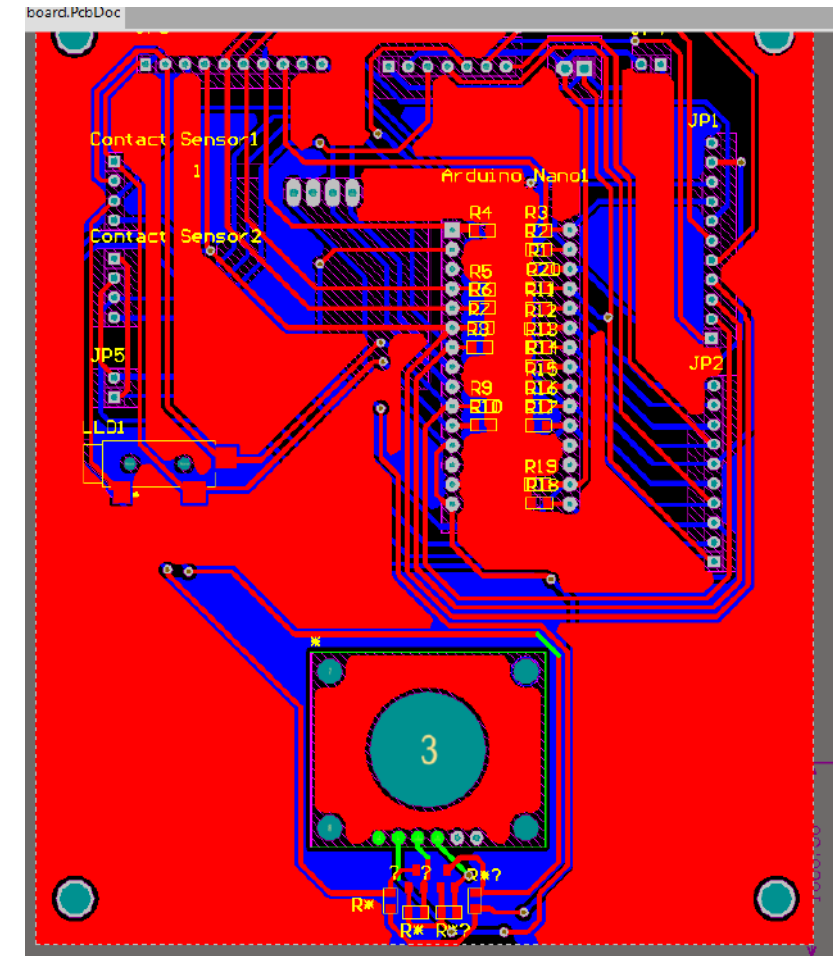
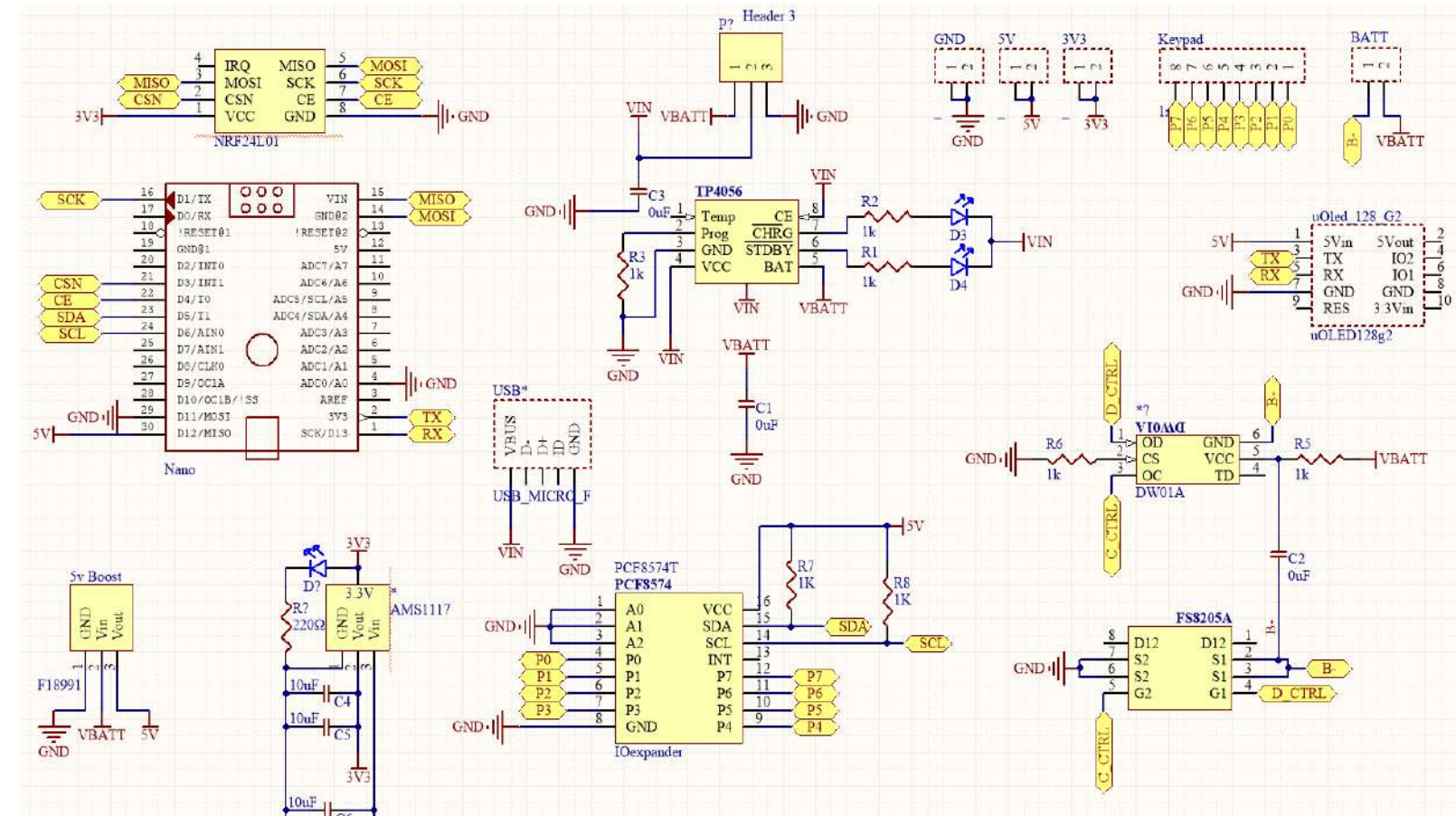
- Based on GX 271 liquid handler
- CO2 dispensing system
 - **Solenoid Valves** controlled by microcontroller/microprocessor
- Custom capping tool
- Laser cut trays to secure bottles



Hardware: Test Stations



Cable Test Station PCB Design



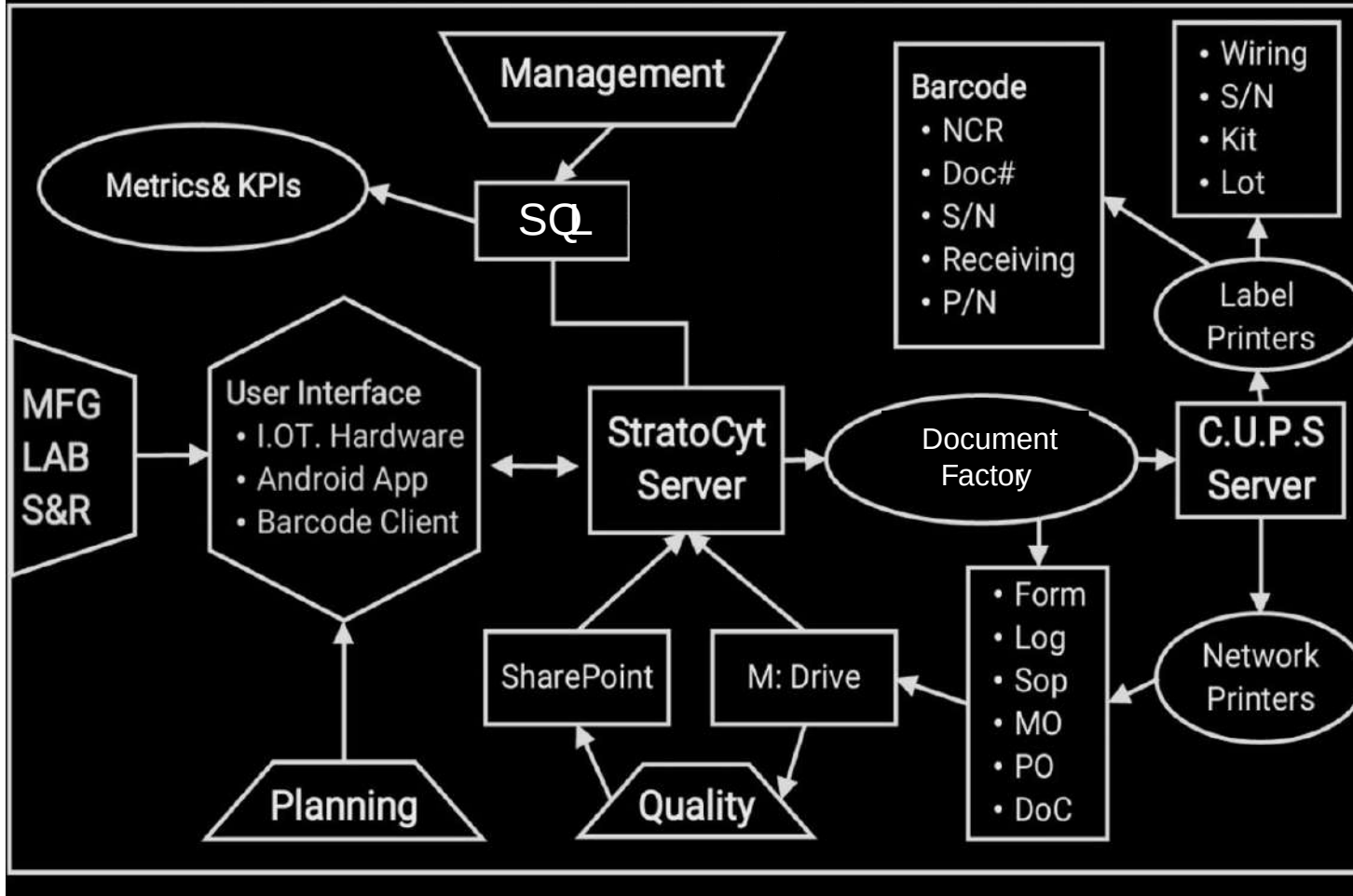
3 & %5 D S L G
(W F K L Q J

3 U R W R W \ S



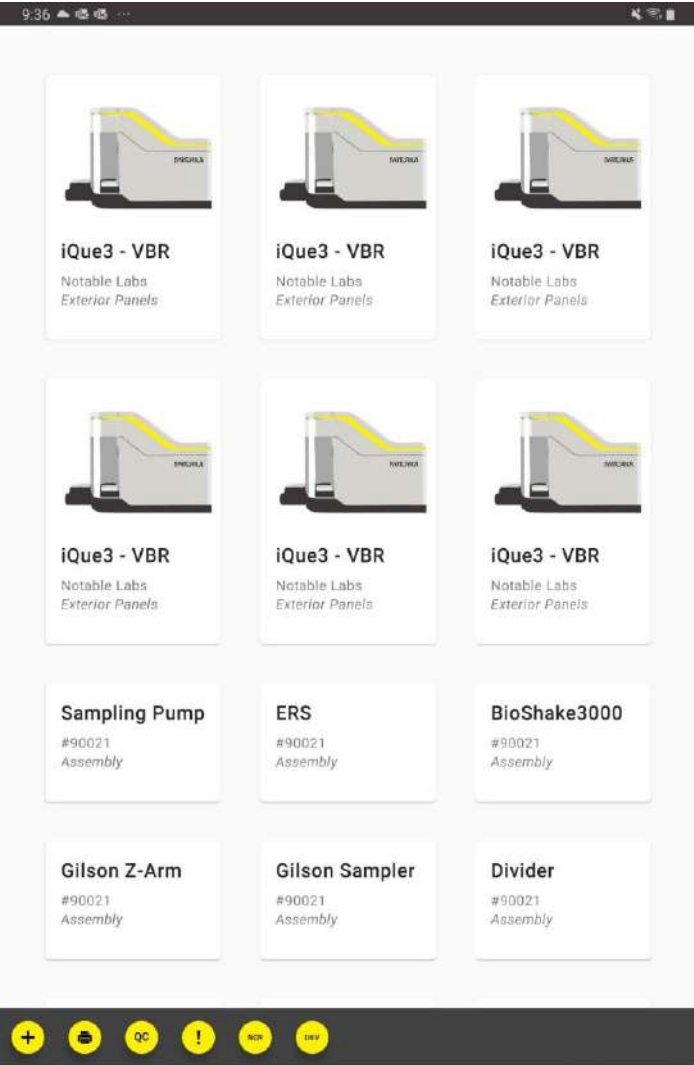
StratoCyt² Custom Automation Server

Reference Diagram



- Microsoft SQL Server Database
 - Stores essential production data for real time metrics
- Python Socket Server
- Automatic document and label generation
 - Automates paperwork system
- Mobile front-end for efficient in process user interface

StratoCyt: Front end Android app

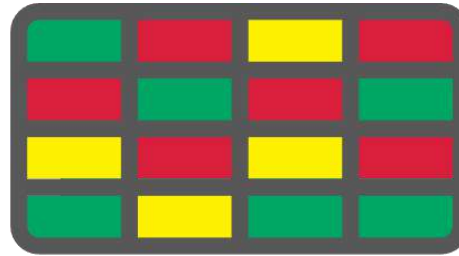


Pick by Light System

Uniquely Illuminable Component Storage

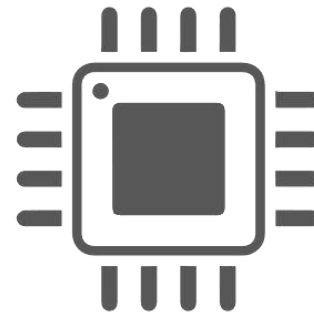
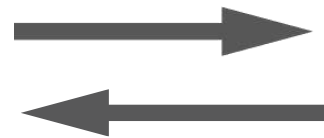
- Less time spent pulling inventory, more during assembly
- Easily avoid shortages via the Kanban system

LED Manifestation of Location/Level
Intuitive Pulling of Inventory



Android UI
(StratoCyt)

Command/Input



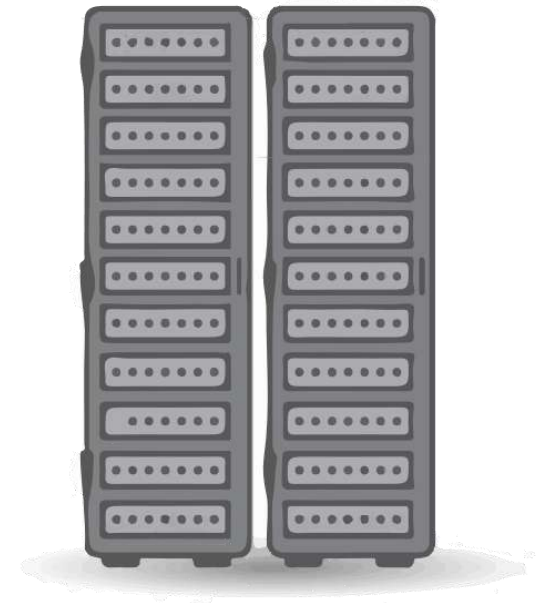
Embedded
Computer

‡ Min/Max
‡ Bin Index



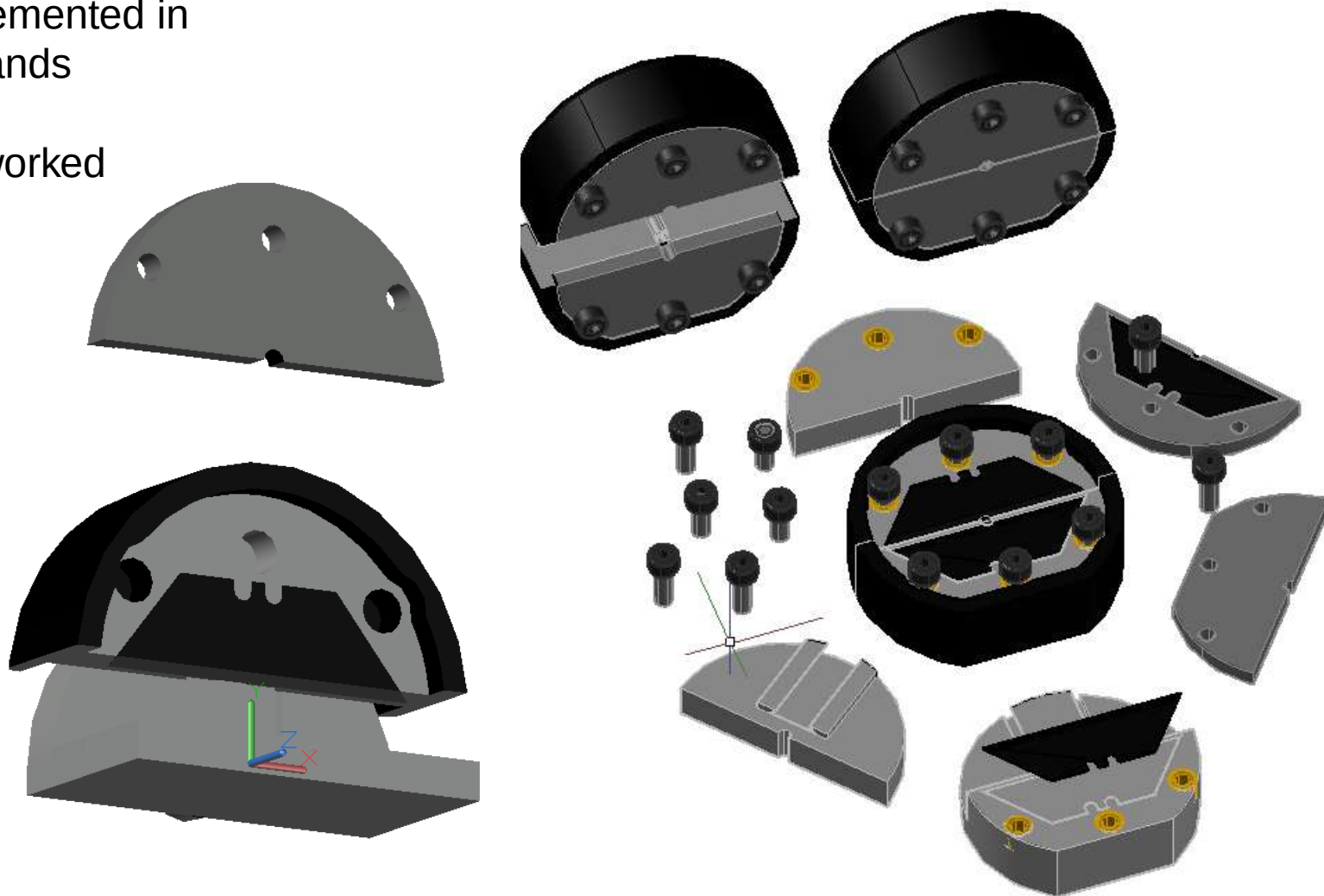
Data Request

SQL Database

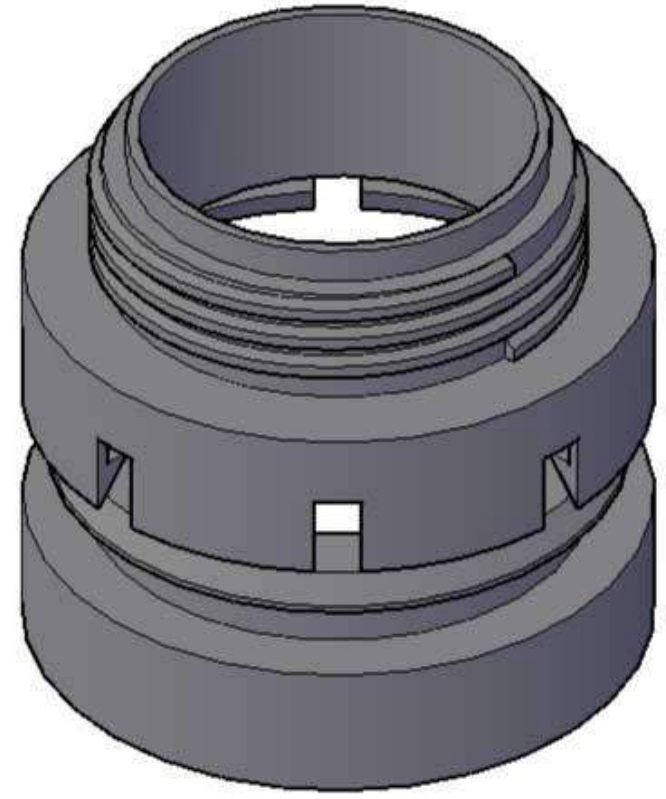
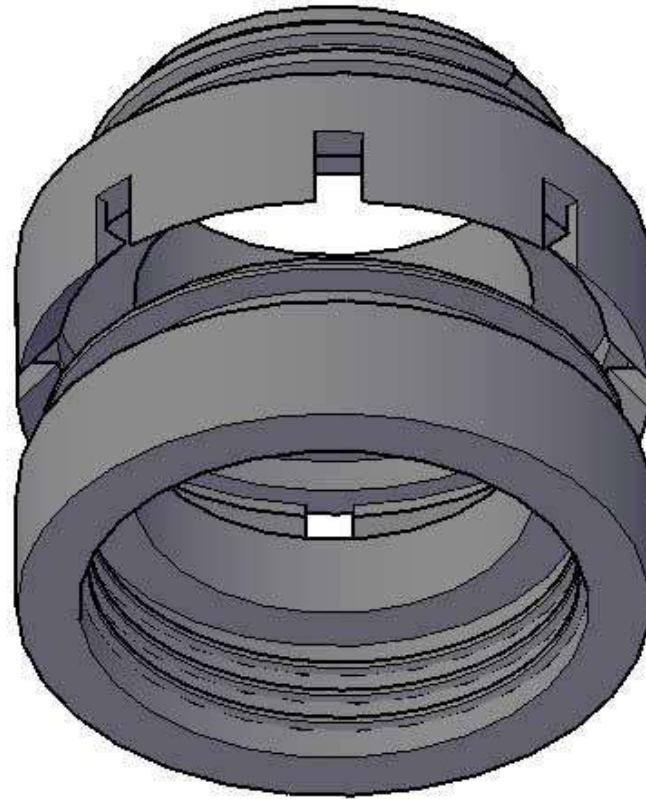
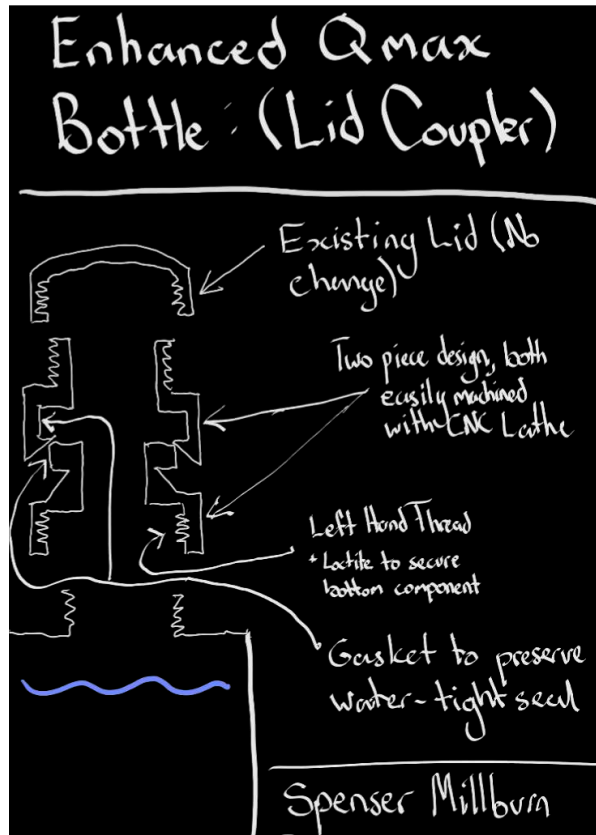


Microfluidics: Rework Jig

- ‡ Designed, fabricated, tested and implemented in 72hr window to meet production demands
- ‡ Effectively recovered USD \$20k in reworked material due to out of warranty vendor error



Qmax² Fluidics/Electrical Bearing`



Load Cell Calibration Jig

